

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Dharyshan B ¹⁹²³¹²¹⁵⁰ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dharyshan

level-4

MD

1) Backtracking search - School Timetable

So: Step-by-step Backtracking

we assign one subject to each period.

Step 1: Assign Math to period 1

$P_1 = \text{Math}$

Constraint checks:-

* Math in $P_1 \rightarrow$ valid

* Math is not in $P_3 \rightarrow$ valid

* No duplicate subject $\rightarrow$ valid

So:-

$P_1 = \text{Math}$

$P_2 = ?$

$P_3 = ?$

Step 2: Assign Science to period 2

$P_2 = \text{Science}$

Constraint checks:-

* Science in $P_2 \rightarrow$ valid

* Science must be before English

* P_2 already has no other subject $\rightarrow$ valid.

So:-

$P_1 = \text{Math}$

$P_2 = \text{Science}$

$P_3 = ?$

Step 3: Assign English to period 3

$P_3 = \text{English}$

Constraint	check
Maths not in P3	Maths is in P1
Science before English	$P2 < P3$
one subject per period	All periods different
English not in P1	English is in P3

Final Timetable:-

period 1 = Maths, period 2 = Science, period 3 = English

2) Uniform Cost Search (UCS) - Foraging Robot

Grid (6x6)

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S . . X . .
. R . X . .
. X . . R .
. . X . X .
R . . . . .
. X . X . G

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Legend:-

S = Start

G = Goal

X = Blocked

R = Risky (cost = 3)

. = Normal (cost = 1)

Step	Expanded node	cost	priority queue
1	S(1,1)	0	(1,2)=1, (2,1)=1
2	(1,2)	1	(2,1)=1, (1,3)=2, (2,2)=4
3	(2,1)	1	(1,3)=2, (3,1)=2, (2,2)=4
4	(1,3)	2	(3,1)=2, (2,3)=3, (2,2)=4
5	(3,1)	2	(2,3)=3, (4,1)=3, (2,2)=4
6	(2,3)	3	(4,1)=3, (3,3)=4, (2,2)=4
7	(4,1)	3	(3,3)=4, (4,2)=4, (5,1)=6
8	(3,3)	4	(4,2)=4, (3,4)=5, (5,1)=6
9	(4,2)	4	(3,4)=5, (5,2)=5, (5,1)=6
10	(3,4)	5	(5,2)=5, (4,4)=6, (3,5)=8
11	(5,2)	5	(5,3)=6, (4,4)=6, (3,5)=8
12	^{5 3} (4,4)	6	(5,4)=7, (4,4)=6, (3,5)=8
13	(4,4)	6	(5,4)=7, (3,5)=8
14	(5,4)	7	(5,5)=8, (3,5)=8
15	(5,5)	8	(6,5)=9, (3,5)=8
16	(6,5)	9	(6,6)=10
17	G(6,6)	10	Goal Reached

optimal path:-

$S \rightarrow (2,1) \rightarrow (3,1) \rightarrow (4,1) \rightarrow (4,2) \rightarrow (5,2) \rightarrow (5,3) \rightarrow (5,4)$
 $\downarrow$
 $(6,5) \leftarrow (5,5)$
 $\downarrow$
 G

Cost calculation :-

* Normal moves $= 10 \times 1 = 10$

* Risky moves $= 0$

* Total cost $= 10$

Final Answer:-

* Optimal path: $S \rightarrow (2,1) \rightarrow (3,1) \rightarrow (4,1) \rightarrow (4,2) \rightarrow (5,2)$

$G_1 \leftarrow (6,5) \leftarrow (5,5) \leftarrow (5,4) \leftarrow (5,3)$

* Minimum Total cost $= 10$.